# CricVision AI

App Wo

Complete guide: setup, architecture, models, tracking,

analysis m

*Branch: stable-clean-ui-8ca838e*

*Commit: 8c*

# Table of Contents



*Tip: Use your PDF reader outline/bookmarks if available.*

## 1. App Overview and Purpose

CricVision AI is a Streamlit-based cricket performance analysis application.

It helps coa

Public MVP scope (branch stable-clean-ui-8ca838e):

- Dashboa

Out of scope on this branch: auto exercise detection, multiplayer, FastAPI backend,

database-b

## 2. How to Run

Prerequisites:

- Python 3

Setup:

cd CricVis

Run the app:

streamlit r

Open the URL shown (default http://localhost:8501). Dashboard loads without

loading hea

Optional Hugging Face fallback (Streamlit Cloud / missing local weights):

- Set HF_

Smoke / dev checks (no models required):

python -m

## 3. User-Facing Features

3.1 Dashboard (Backends/src/ui/dashboard.py)

- Quick ac

3.2 Live Session (Backends/src/ui/live_session.py)

- Webcam

3.3 Video Analysis (Backends/src/ui/video_analysis.py)

- Upload r

3.4 Session Results (Backends/src/ui/results_page.py)

- Loads da

# 4. Architecture and Data Flow

Layered architecture:

[Streamlit UI]  main.py -> dashboard | live_session | video_analysis | results

|

Text data-flow (uploaded video):

User uploads clip

-> vide

Key design rule: one shared detection timeline feeds all reports.

No duplica

## 5. File and Folder Map

Entry and UI:

main.py

Video pipeline:

video_pip

Analysis and agents:

analysis/s

Calibration (model-free, deterministic):

calibration

Models and config:

models/m

Storage and outputs:

storage/s

## 6. Models Used Where

Registry keys (models/model_registry.py):

current_best (default)

Path: Mo

cricshot_ball

Path: Mo

cricshot_bat

Path: Mo

player_type (experimental, not wired)

striker_se

Legacy weights kept on disk (ensemble/advanced only):

Models/ba

Loading behavior:

- No YOL

## 7. Analysis Modes and Settings

7.1 Smart speed modes (video upload, default path)

Smart Bal

7.2 Analysis type (Advanced expander)

Bowling A

7.3 Detection presets (config/constants.py)

Fast Bowl

7.4 Overlay and output toggles

Generate

7.5 Live Session presets

Same DE

## 8. Tracking Approach on This Branch

This branch (stable-clean-ui-8ca838e) uses deterministic 2D tracking repair,

not true 3D

Video Analysis pipeline:

1. YOLO p

Live Session (separate path):

- BallKalm

Constants (config/constants.py):

LOW_CO

Explicit non-goals on this branch:

- No metri

## 9. Data Outputs and Storage Paths

Session persistence:

data/sess

Generated artifacts (per analysis run):

outputs/pr

Remote model cache:

Models/re

Session record fields (normalized by session_store.py):

id, createe

Note: no automatic retention policy; outputs accumulate until manual cleanup.

Do not dele

## 10. Branch Context: stable-clean-ui vs main

Current branch: stable-clean-ui-8ca838e (commit 8ca838e)

Theme: Cl

Recent focus on this branch:

- Tabbed

Compared to experimental/main tracking work:

- This bra

## 11. Limitations and Known Gaps

Accuracy and environment:

- 2D pixel

Architecture / maintenance:

- video_ar

Models:

- shot_cla

Features not in production:

- Wagon v

## 12. Suggested User Workflow (Step-by-Step Planner)

Phase A - Setup (one time)

1. Clone r

Phase B - Calibrate practice context (per session)

1. Open V

Phase C - Analyze a delivery

1. Upload

Phase D - Review and save

1. Check

Phase E - Live practice (optional)

1. Open L

Phase F - Maintenance

1. Periodi

## Document Info

Title: CricVision AI - App Working Planner

Branch: sta